

MERIDIAN FAMILY OFFICE · COPILOT v10

# AI Copilot — Test Suite & Results

Verification of the deterministic engine, the deliverable layer, and the end-to-end app

**135/135 tests passed · 16 July 2026**

*Runs fully offline in DEMO\_MODE — no API key, no network, no billing.*

## Summary

The suite verifies one central invariant: **every number is computed deterministically; documents and instructions are advisory context that shape prose, never figures.**

| Layer       | What it proves                                                    | Tests |
|-------------|-------------------------------------------------------------------|-------|
| Engine      | Parsing, reconciliation, suitability policy, grounded Q&A         | 30    |
| Deliverable | Document ingestion, portable prompt, narrative, deck render       | 23    |
| App         | The Streamlit app end-to-end across every view                    | 19    |
| Business    | Client macro views (rates, war, tax) shape prose, not the numbers | 63    |

## Statement ingestion · 9 tests

**Engine** Parse the three custodian formats, reconcile against each statement's own total, convert FX to USD, and keep a PII-free continuity key. Every downstream number rides here.

file: tests/test\_statement\_parser.py

| # | Scenario verified                                                                                                                                      | Result |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 1 | Parse custodian A (USD, per-share CSV) into valued positions.                                                                                          | PASS   |
| 2 | Semicolon CSV, European decimals, EUR/CHF → converted to a USD base.                                                                                   | PASS   |
| 3 | JSON custodian carries illiquid alternatives + a Lombard loan (a liability).                                                                           | PASS   |
| 4 | One sample deliberately fails its checksum by ~\$436 — it must be flagged, not silently trusted.                                                       | PASS   |
| 5 | The ~\$436 break on custodian B reconciles under a generous tolerance but fails under a tight one — proving the checksum knob actually gates the flag. | PASS   |
| 6 | Continuity key is deterministic across re-parses and carries no PII.                                                                                   | PASS   |
| 7 | Continuity check compares month-to-month keys.                                                                                                         | PASS   |
| 8 | Consolidate all three custodians into one holdings list.                                                                                               | PASS   |
| 9 | gross – debt = net, and gross is the sum of position base values.                                                                                      | PASS   |

## Suitability & policy · 12 tests

**Engine** The deterministic policy checks. The mandate decides what a breach MEANS; plus concentration, liquidity, unhedged-FX, hard regulatory blocks and instruction-diff.

file: tests/test\_suitability.py

| # | Scenario verified                                                                                                             | Result |
|---|-------------------------------------------------------------------------------------------------------------------------------|--------|
| 1 | A single overweight equity book, evaluated under each mandate, yields the mandate-appropriate enforcement on the band breach. | PASS   |
| 2 | A balanced book inside every band produces no actionable breach.                                                              | PASS   |
| 3 | A single stock above the 10% single-security cap is caught; a diversified fund of the same weight is not.                     | PASS   |
| 4 | Parked liquidity (money-market) is exempt from the single-position cap.                                                       | PASS   |
| 5 | Below the min-liquid floor (here mostly illiquid alternatives) is flagged.                                                    | PASS   |
| 6 | Economic FX exposure above the max-unhedged cap is flagged.                                                                   | PASS   |
| 7 | Sanctioned instrument hard blocks even execution only.                                                                        | PASS   |
| 8 | Complex product appropriateness blocks when not allowed.                                                                      | PASS   |

|    |                                                                                                                               |      |
|----|-------------------------------------------------------------------------------------------------------------------------------|------|
| 9  | Client asked for a treasury sleeve; the proposal holds a corporate-credit bond fund of the wrong duration → a deviation flag. | PASS |
| 10 | Instruction diff satisfied when matched.                                                                                      | PASS |
| 11 | Worst enforcement ranking.                                                                                                    | PASS |
| 12 | The three-custodian sample book, under its advisory profile, lands on a 'flag' gate (bands breached, nothing hard-blocked).   | PASS |

## Conversational Q&A · 9 tests

**Engine** *Routing picks the right deterministic tool from the phrasing; answers stay grounded in computed facts; the router never crashes on odd input.*

file: tests/test\_portfolio\_qa.py

| # | Scenario verified                                                       | Result |
|---|-------------------------------------------------------------------------|--------|
| 1 | 'Net worth' routes to the totals tool.                                  | PASS   |
| 2 | An FX / currency question routes to the currency tool.                  | PASS   |
| 3 | A suitability / mandate question routes to the suitability tool.        | PASS   |
| 4 | A 'largest position' question routes to the concentration tool.         | PASS   |
| 5 | A liquidity question routes to the liquidity tool.                      | PASS   |
| 6 | A reconciliation question routes to the data-quality tool.              | PASS   |
| 7 | Every answer carries facts tied to the computed book.                   | PASS   |
| 8 | The deterministic briefing fact-sheet spans every domain.               | PASS   |
| 9 | The router must always resolve to SOME tool — never crash on odd input. | PASS   |

## Document ingestion · 7 tests

**Deliverable** *The research / other uploaders read text deterministically so content can be folded into the prompt as advisory context — never a source of figures.*

file: tests/test\_doc\_extract.py

| # | Scenario verified                                                         | Result |
|---|---------------------------------------------------------------------------|--------|
| 1 | Plain text and Markdown read cleanly.                                     | PASS   |
| 2 | HTML is stripped to readable text (tags removed).                         | PASS   |
| 3 | An .eml email body is extracted.                                          | PASS   |
| 4 | CSV and JSON pass through as text.                                        | PASS   |
| 5 | An unsupported file type fails gracefully (no crash).                     | PASS   |
| 6 | Very large documents are truncated (MAX_CHARS) and marked, never dropped. | PASS   |
| 7 | The extraction result always has the same shape.                          | PASS   |

## Prompt & narrative · 10 tests

**Deliverable** *The portable prompt bundles every input; the keyless commentary REFLECTS the documents (the v10 fix) while only ever quoting the computed numbers.*

file: tests/test\_narrative.py

| # | Scenario verified                                                         | Result |
|---|---------------------------------------------------------------------------|--------|
| 1 | The prompt carries every block (FACTS, intake, holdings, docs, tactical). | PASS   |
| 2 | Document text is embedded verbatim in the prompt.                         | PASS   |
| 3 | Without documents, that content is absent from the prompt.                | PASS   |
| 4 | The FACTS block is valid JSON and holds the numbers.                      | PASS   |
| 5 | v10 fix: documents change the deterministic commentary.                   | PASS   |

|    |                                                                                                                                                                                                     |      |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 6  | The commentary quotes short excerpts of the tactical text and docs.                                                                                                                                 | PASS |
| 7  | The keyless summary must use the model's own numbers verbatim and never invent a figure that isn't already somewhere in the computed model.                                                         | PASS |
| 8  | The commentary carries the 'not investment advice' disclaimer.                                                                                                                                      | PASS |
| 9  | The prompt that generates the commentary slide must NOT ask the model to build a deck / produce files — otherwise the model returns the slide-by-slide spec, which leaks onto the commentary slide. | PASS |
| 10 | The portable whole-deck prompt (a separate artifact) still asks for the full deck — that behaviour is intentional and must stay distinct from the commentary.                                       | PASS |

## Proposal deck render • 6 tests

**Deliverable** *HTML / PPTX / PDF renderers agree, figures are copied from the model, and documents change every exported format.*

file: tests/test\_generate\_proposal.py

| # | Scenario verified                                               | Result |
|---|-----------------------------------------------------------------|--------|
| 1 | All three renderers (HTML/PPTX/PDF) produce valid output.       | PASS   |
| 2 | The deck shows the computed net-worth & gross figures.          | PASS   |
| 3 | The commentary slide appears only when a narrative is set.      | PASS   |
| 4 | v10 fix: documents change the HTML, PPTX and PDF exports.       | PASS   |
| 5 | With documents, the deck surfaces the client's liquidity event. | PASS   |
| 6 | The deck carries the 'not independently verified' disclaimer.   | PASS   |

## End-to-end app • 19 tests

**App** *Drive the real Streamlit app across the views a banker walks through: load samples, review each view, change the mandate & tactical instructions — and never error.*

file: tests/test\_app\_scenarios.py

| #  | Scenario verified                                                                                                                                                       | Result |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 1  | The app boots with no book loaded.                                                                                                                                      | PASS   |
| 2  | Each sample statement loads into the Overview view. (3 variants)                                                                                                        | PASS   |
| 3  | Overview / Holdings / Suitability / Proposal all render. (4 variants)                                                                                                   | PASS   |
| 4  | Proposal does not auto generate commentary.                                                                                                                             | PASS   |
| 5  | Generate button produces commentary on click.                                                                                                                           | PASS   |
| 6  | No overlay → no tilt banner on the proposal.                                                                                                                            | PASS   |
| 7  | Selecting a macro overlay tilts the target the rebalance chases, and the Proposal view says so — the up-sleeve's target rises and the down-sleeve's falls. (2 variants) | PASS   |
| 8  | The Proposal view shows the assembled prompt and commentary.                                                                                                            | PASS   |
| 9  | Tactical instructions flow into the on-screen prompt.                                                                                                                   | PASS   |
| 10 | Under discretionary, band breaches HARD-BLOCK; under advisory they only flag. The same book therefore reads differently on the Suitability view. (2 variants)           | PASS   |
| 11 | A single-custodian book still produces a proposal.                                                                                                                      | PASS   |
| 12 | The Proposal view with no book degrades gracefully.                                                                                                                     | PASS   |

## Business scenarios • 39 tests

**Business** *What happens to the output on a client macro view — a rate hike/cut, a war (or its end) prompting alternatives, a tax rise/cut: it reaches the AI, shapes the commentary, and never moves the grounded figures.*

file: tests/test\_business\_scenarios.py

| # | Scenario verified                                                                                                                                                  | Result |
|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 1 | A client macro view (rate/war/tax) reaches the AI verbatim. (12 variants)                                                                                          | PASS   |
| 2 | The macro view shapes the advisory commentary. (12 variants)                                                                                                       | PASS   |
| 3 | A macro opinion shapes prose, never figures: the FACTS block (every number the deck shows) is byte-for-byte identical with and without the scenario. (12 variants) | PASS   |
| 4 | 'What if rates are hiked?' → the deterministic tool states the fixed-income exposure but refuses to estimate a +100bp impact it cannot source.                     | PASS   |
| 5 | A rate-move question routes to the rate tool.                                                                                                                      | PASS   |
| 6 | The scenario moves the deck's prose, not its tables.                                                                                                               | PASS   |

## Macro overlays · 24 tests

**Business** A client macro view MOVES the recommendation: each overlay is a fixed, bounded, provenanced tilt of the target allocation — the rebalancer chases it, so trades shift in the scenario's direction while every figure stays deterministic.

file: tests/test\_macro\_overlay.py

| # | Scenario verified                                                                                                                                    | Result |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 1 | Each overlay moves the target sleeves in the documented direction. (5 variants)                                                                      | PASS   |
| 2 | A tilt sums to zero — the book stays fully invested. (5 variants)                                                                                    | PASS   |
| 3 | No sleeve is driven negative by a tilt. (5 variants)                                                                                                 | PASS   |
| 4 | 'No overlay' changes nothing.                                                                                                                        | PASS   |
| 5 | An unknown overlay falls back to a no-op.                                                                                                            | PASS   |
| 6 | Reported changes name only moved sleeves; the summary is provenanced. (5 variants)                                                                   | PASS   |
| 7 | Tilts are modest: no sleeve moves more than 8pp from its strategic target, so a tilt never breaches a normal $\pm 15\%$ suitability band on its own. | PASS   |
| 8 | If a tilt would drive a sleeve negative, it clamps at 0 and renormalises so the total is preserved (no phantom allocation).                          | PASS   |

## How to run

```
python3 -m pytest                                # whole suite
```

```
python3 -m pytest tests/test_suitability.py -v
```

```
python3 -m pytest -o addopts='' -v                # per-test PASS lines
```